

A Review on the Interface Design of Automotive Head-Up Displays for Communicating Safety-Related Information

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Automotive head-up displays (HUDs) have the advantage of allowing drivers to keep their eyes forward while driving by superimposing visual information on top of the drivers' forward field of view (FoV). In order to provide the intended advantage, HUDs must be designed such that they respect the characteristics and capacities of the human information processing system and accommodate the specific contexts of information use. Poorly designed HUDs indeed can adversely affect driving safety by creating new sets of problems, including visual clutter, information overload, inattentive blindness and cognitive capture (Gish and Staplin, 1995; Pauzie, 2015; Tufano, 1997; Ward and Parkes, 1994).

During the last few decades, various research studies have proposed different HUD displays that present HUD information in particular styles. However, it is not well understood what type of display would be most advantageous or adequate for effectively communicating each information type and thus best serve drivers in performing the associated driving task. Relatively little research has been conducted to evaluate the available HUD displays in the interface design.

As an initial effort towards addressing the knowledge gap, the objective of the current study was to provide a review of the existing HUD displays focusing on the interface design. Among the variety of HUD information types, the current review was intended to cover different types of safety information. For each type of safety information, the HUD displays proposed by the automobile industry and academic research were examined, in terms of their structures and behaviors and also related human factors display design principles. On the basis of the review results, this study suggests some future research directions that would help develop useful and effective HUD displays.

This study conducted two literature searches, one for documents describing existing commercial HUD displays, and the other one for research articles proposing or evaluating automotive HUD displays for communicating safety-related information. Thirteen major automobile manufacturers providing HUD systems, such as Audi, BMW Group, Ford, General Motors, Honda, Hyundai/KIA, Jaguar Land Rover, Mercedes-Benz, PSA Peugeot Citroen, Renault, SAAB, Toyota, and Volvo, were considered. The details of the interface designs of the manufacturers' HUD displays were examined using the information provided in the vehicle manuals. Research articles were searched by utilizing terms describing four concepts: (1) head-up display (head up display(s), head-up display(s), HUD(s)), (2) automobiles (automotive, vehicle(s), car(s), automobile), (3) interface design (display, design, interface, augmented reality, human factors, system), and (4) safety information (safety, warning,

alert). As a result, a total of 15 studies were included in this review.

The review results indicated that safety-related HUD displays proposed by academic research studies were mostly AR-based and contact-analog. On the other hand, all of the commercial safety-related HUD displays were of the unregistered type and did not utilize the AR technology.

It is not clear why the existing commercial HUD displays did not adopt the AR technology. Perhaps, it may be due to some technological challenges in incorporating the AR technology into the automotive HUD system. Alternatively, it may be that the efficacy of the AR HUD technology has not been confirmed for creating safety-related HUD displays.

This study examined the HUD displays in terms of the human factors display design principles (Wickens et al., 2003). Many of the proposed displays indeed were based on some of the well-known display design principles, such as the principles of proximity compatibility, information access cost minimization, predictive aiding, color coding and consistency.

However, the display proposed by George et al. (2012) seemed to provide a relatively large amount of information in one display. In this case, the legibility of the display may decrease and cause confusion in terms of discriminability. In addition, visual complexity may increase, which may degrade information processing. In dangerous situations, displays should be simple and should not require too much perception or interpretation. Future research will need to determine to what degree the complexity of the display is acceptable in hazardous or safety-critical driving situations. Research is also needed to identify the individual differences in the acceptance levels of visual complexity.

On the basis of the review results, some future research directions were identified:

- Research should attempt to develop design principles/guidelines that help designers identify an appropriate user interface type when given an information characteristic and its usage context.
 - What are the information characteristics suitable for contact-analog and unregistered display formats? Which of the two display formats would be more effective, under various circumstances, especially in situations where nearby hazards must be detected quickly?
- Research is needed to investigate how to design and evaluate HUDs taking into account the drivers' information processing capabilities under safety-critical driving situations.
 - How many HUD displays can be presented without exceeding the drivers' information

processing capabilities under safety-critical driving situations? In this regard, what are the priority levels of different HUD displays and how can they be determined?

- What is the acceptable level of visual complexity of a single or multiple displays within the drivers' information processing capabilities? What are the individual differences in the acceptance levels of visual complexity?

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